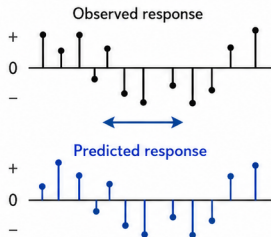


1 Direction recovery

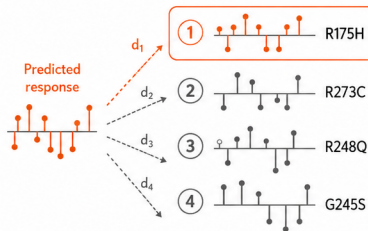


Pearson- δ
 δ -cosine



Does the predicted response point in the correct direction?

2 Allele discrimination



PDS

$d(\text{pred, own}) < d(\text{pred, others})$



Does the prediction identify its own allele among held-out variants?

defining allele-level requirement

3 DE fidelity

Observed (changed genes)			Predicted (changed genes)		
Gene A	↑	✓	Gene A	↑	
Gene B	↓	✓	Gene B	↓	
Gene C	↑	✓	Gene C	↑	
Gene D	↓	-	Gene X	↓	
Gene E	↑	✓	Gene E	↑	
Gene F	↓	-	Gene Y	↓	
Gene G	↑	-	Gene Z	↑	
Gene H	↑	-	Gene H	↓	

- DE overlap
- DE-LFC rank correlation
- direction agreement



Which genes change, by how much, and in which direction?

increasing evaluation resolution



shared response direction



allele identity



variant-specific gene changes



Direction recovery is necessary, but allele discrimination defines success for held-out variant prediction.